

Data Converters

Exercise 07 – Pipeline ADC

Task I – Analysis of a Pipeline ADC

The block diagram of a 3-stage pipeline ADC is shown in Figure 1.

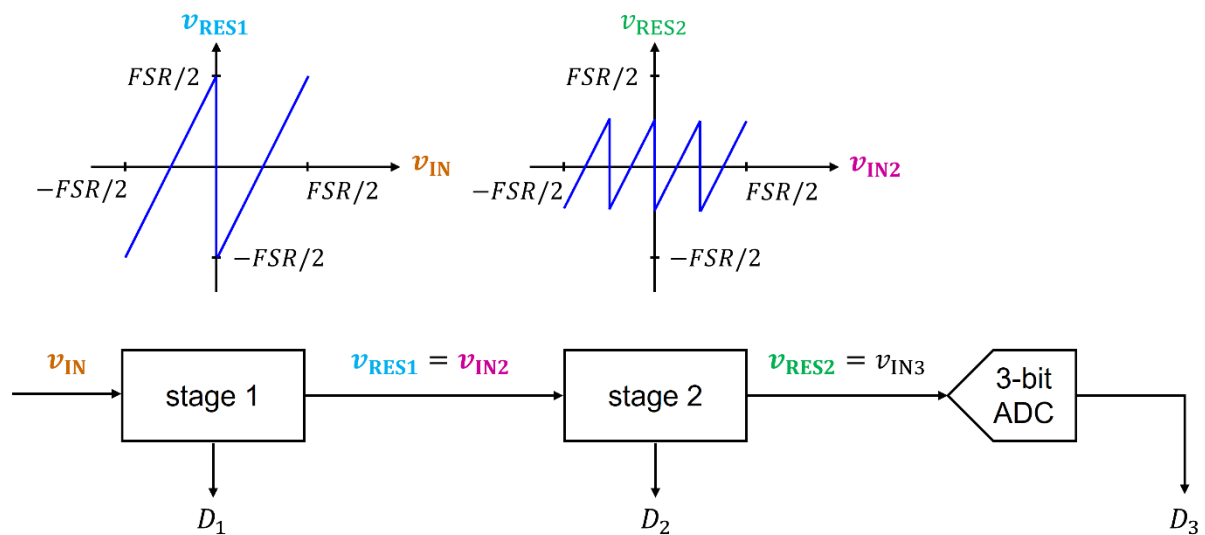


Figure 1: Pipeline ADC.

- Determine the resolution of the sub-ADC of the first and the second stage.
- Determine $v_{RES1}(v_{IN}, e_{q1})$ $v_{RES2}(v_{IN2}, e_{q2})$, where q_1 and q_2 are the quantization error of the sub-ADC of the first and the second stage.
- Determine the gain G_1 and G_2 of the first and the second stage.
- Determine the resolution of the pipeline ADC.
- Compare the resolution of the pipeline ADC with the sum of the resolutions of its sub-stages. What is the explanation for the inconsistency?
- Determine the maximum tolerable error of the sub-ADC of the first and the second stage.
hint: The input signal of the final ADC must be limited to its input full-scale range.
- Considering the results of task I.f, briefly explain why the ADC is still functional for

$$-FSR/2 - FSR/16 \leq v_{IN} \leq FSR/2 + FSR/16$$

and

$$-FSR/2 - FSR/8 \leq v_{IN2} \leq FSR/2 + FSR/8$$

although *stage 1* and *stage 2* are overloaded.

- h. Consider Figure 1 and derive the transfer characteristic of the ADCs used in *stage 1* and *stage 2*, i.e., D_1 vs. v_{IN} and D_2 vs. v_{IN2} .
- i. Use your results of task I.b, task I.h as well as the transfer characteristic of the 3-bit ADC given in Figure 2 to determine D_1 , D_2 , D_3 and $D_{OUT} = D_1 + D_2/G_1 + D_3/G_1G_2$ in the analog domain for $v_{IN} = 0.126 FSR = 63/500 FSR$. Moreover, determine v_{RES1} and v_{RES2} .

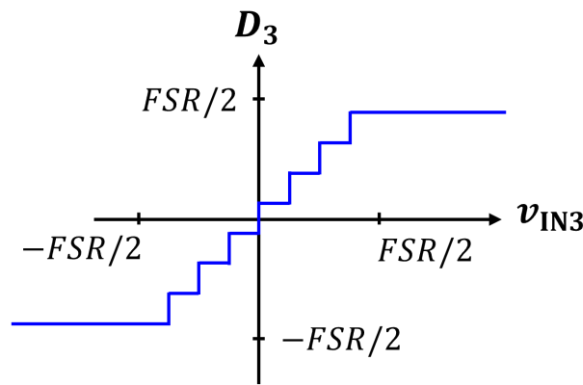


Figure 2: Transfer characteristic of the 3-bit ADC.

- j. Repeat task I.i using the transfer characteristic for the ADC of *stage 2* shown in Figure 3 where the transition at $-FSR/4$ is shifted to $-FSR/8$.

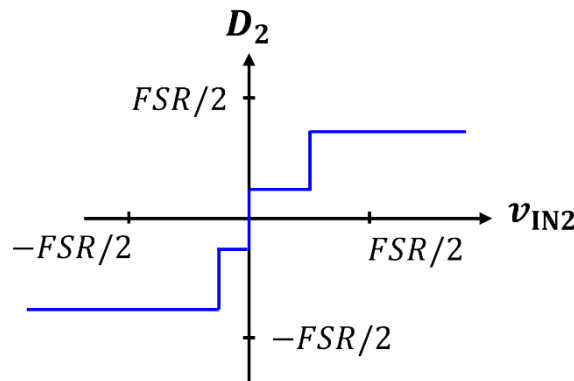


Figure 3: Non-ideal transfer characteristic of the ADC of stage 2.

- k. Repeat task I.i using the transfer characteristic for the DAC of stage 2 shown in Figure 4 where the analog output for code 01 is $-FSR/16$ instead of $-FSR/8$.

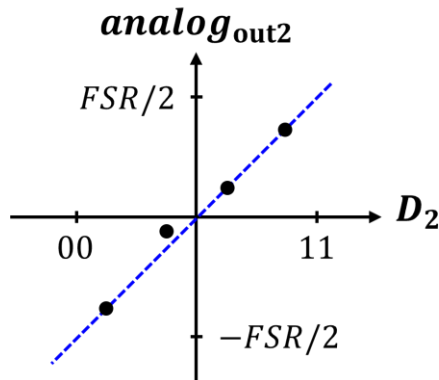


Figure 4: Non-ideal transfer characteristic of the DAC of stage 2.

- l. Comparing your results of task I.j with task I.k: Briefly explain why the pipeline ADC can compensate for the error of the ADC but not of the DAC in stage 2.

Task II – Yet Another Analysis of a Pipeline ADC

The block diagram of an arbitrary pipeline ADC is shown in Figure 5. Unfortunately, the interstage gain between the first stage and the remaining stages summarized in the block backend was implemented as $3/2$ instead of 2 .

The 2 – bit ADC has transition levels at $-FSR/4$, 0 , and $FSR/4$. Moreover, the 2 – bit DAC and backend are ideal and the backend can be assumed to have infinite resolution/precision.

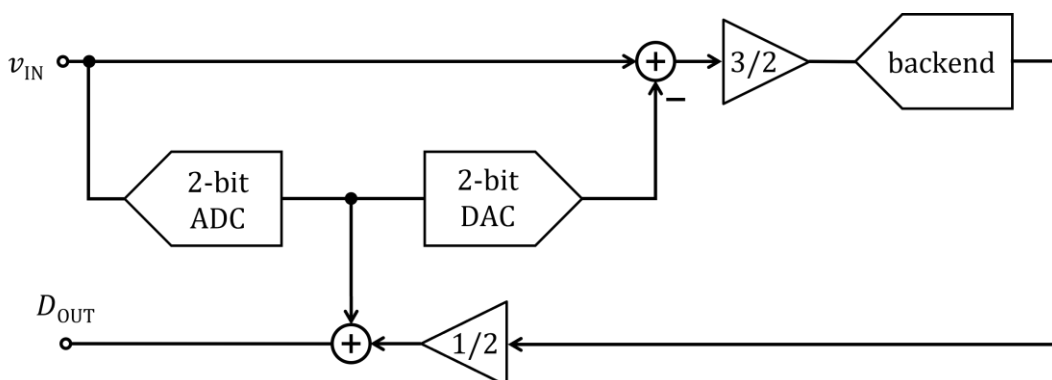


Figure 5: Arbitrary pipeline ADC.

- Draw the linearized model of the pipeline ADC.
- Determine D_{OUT} in the analog domain.

- c. Sketch D_{OUT} versus v_{IN} in the analog domain for one segment of the 2 – bit ADC, e.g., between 0 and $FSR/4$.
- d. How large is the slope of the segment?
- e. How large is the jump at the transition points?

Task III – Final Analysis of a Pipeline ADC

A 9-stage pipeline ADC uses a 2-bit flash ADC in the last stage. All other stages are implemented using a circuit block that generates a residue according to Figure 6. All interstage gains are set to $\times 2$.

- a. Briefly explain why this circuit block is called a 1.5-bit stage.
- b. Determine the maximum tolerable error of the sub-ADCs.
- c. A friend of yours comes up with an idea that he/she considers very clever: In order to save power, all the interstage gains of $\times 2$ were removed. Instead, the full-scale ranges were scaled down by a factor $\times 1/2$ from stage to stage, i.e., $FSR/2$, $FSR/4$, $FSR/8$, and so on.

Which of the 1.5-bit stages thus becomes most sensitive to sub-ADC errors?

Determine the maximum tolerable error of the sub-ADCs.

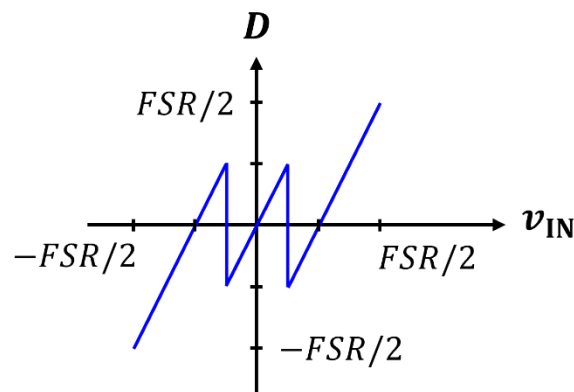


Figure 6: Circuit block used in the pipeline ADC of task III.